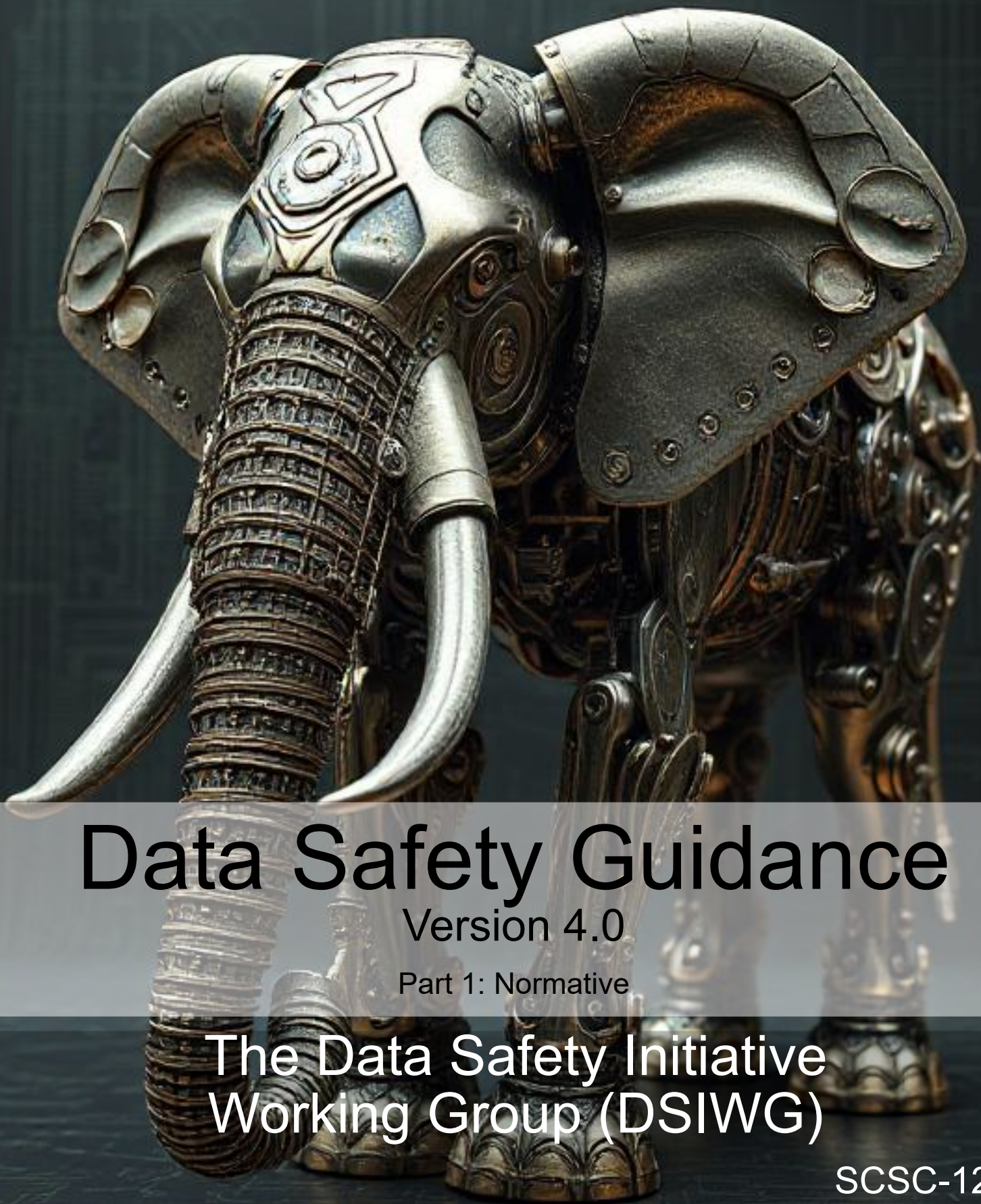




Data Safety Guidance

Version 4.0

Part 1: Normative

The Data Safety Initiative
Working Group (DSIWG)

SCSC-127k

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Completely
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front
matter

The [Safety-Critical Systems Club \(SCSC\)](#) is the professional network for sharing knowledge regarding safety-critical systems. It brings together:

- engineers and specialists from a range of disciplines working on safety-critical systems in a wide variety of industries;
- academics researching the arena of safety-critical systems;
- providers of the tools and services that are needed to develop the systems; and
- the regulators who oversee safety.

Through publications, seminars, workshops, tutorials, a web site and, most importantly, at the annual [Safety-critical Systems Symposium \(SSS\)](#), it provides opportunities for these people to network and benefit from each other's experience in working hard at the accidents that don't happen. It focuses on current and emerging practices in safety engineering, software engineering, and product and process safety standards.

This document was written by the [Data Safety Initiative Working Group \(DSIWG\)](#), which is convened under the auspices of the [SCSC](#). The document supports the [DSIWG's](#) vision, which is to have clear guidance that reflects emerging best practice on how data (as distinct from software and hardware) should be managed in a safety-related context. This update takes account of the consensus that a process-based guidance document will complement existing safety management processes, making it more usable. It was formally released at [SSS'25, 4–6 February 2025](#), details of which may be found at <https://scsc.uk/e1099>

Update
publication
event

Comments on this document are actively encouraged. These can be emailed to:

comments@data-safety.scsc.uk.

Alternatively, a comments submission form is available at:

data-safety.scsc.uk/comments.

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Data Safety Guidance

The Data Safety Initiative Working Group [DSIWG]

February 2026

Add SSS'26 dates and publication details

Change History

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4.0	The DSIWG Team	For publication at SSS'26	10-FEB-2026

Update this whole section

Changes Since the Last Edition

This is a major rewrite of the guidance. The document has been split into three volumes , each giving progressively more detail.

The process remains substantially the same. The intention is that compliance with this version does imply compliance with earlier versions, but some informative material has been made normative so compliance with earlier versions of the guidance does not guarantee compliance with this version.

Update future work

Future work

MCA Ltd has continued to work with the [DSIWG](#) to develop a prototype software tool to assist in the automation of the processes described in this guidance document. A working version of the tool has been developed and organizations that could benefit from the use and further development of the tool are urged to contact MCA at [Mission Critical Applications Limited \(radish@mca-ltd.com\)](mailto:radish@mca-ltd.com).

A number of improvements to the guidance are currently planned. These improvements are intended to clarify the application of the data safety process and include:

- further detail on the assurance of communications and data flows,
- data safety considerations associated with distributed [datasets](#) and Blockchain,
- addition of new [treatments](#) to the tables in **??**,
- review of the tables of [treatments](#), with the aim of making them easier to use,
- further explanation of some [treatments](#), where their use or benefit is not immediately apparent,

Consider whether future work needs to be included in all parts

Review future work

- reordering of parts of the document to improve readability, especially as regards likelihood,
- further detail on tool assurance,
- harmonisation of language and guidance on how organizations may expand the tables to incorporate their own internal processes.
- guidance on the application of the data safety culture questionnaire,

Several of these changes are likely to cause parts of the document to be re-ordered – they have therefore been deferred to the next major update, in version 4.0 of the guidance.

If you or your organization are interested in learning more about the work of the [DSIWG](#) or joining any of the sub-groups, please visit the [SCSC](#) website, where more information including contact details may be found on the “Working groups” section of the site.

Related working groups

The [SCSC](#) sponsors initiatives to develop methods and techniques through a number of working groups. These groups each address safety aspects peculiar to their domain, including data aspects when appropriate. The current list of working groups includes:

- Assurance Cases,
- Security Informed Safety,
- Safe AI Working Group,
- Safer Complex Systems.

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Foreword

Data is here. Data is growing. Data is causing harm.

Data is here: Data is becoming ever more important in our lives: influencing, managing and even controlling many critical aspects. The use of [artificial intelligence \(AI\)](#) systems is a new, exciting, but potentially hazardous use of data. [Large language model \(LLM\)](#) based systems are trained on vast amounts of data, and it is this data which enables them to be useful.

Some of this data is related to our personal safety and well-being. Consider, for example, the importance of data defining the layout of railway signals, data that indicates the position of underwater obstructions in nautical channels or data that is used to train a vision recognition system to detect tumours in medical images. Organizations now make significant decisions (including safety-related decisions) based solely on data held in systems. Hence, organizations need to safely manage, control and process their data. In particular, they must actively manage key [data properties](#) that preserve safety.

Data is growing: There are at least two reasons why the use of data has grown and, equally important, why it is expected to continue to grow. The first relates to the rapid expansion of the area loosely termed “Big Data”, including the use of large [datasets](#) to support machine learning and [AI](#) applications. The second is the growing use of systems of systems, where data is the lifeblood that connects together disparate elements and allows a cohesive capability to be built. Put simply, the need to address data-related issues is a pressing problem and will continue to be so.

Data is causing harm: Strictly speaking, data can neither cause nor prevent harm. However, mistakes in data, or the inappropriate use of data, within safety-related systems have been factors in a number of documented accidents and incidents. Examples include aircraft attempting to take off from the wrong runway (and consequently crashing), ships running aground, and patients being exposed to higher than planned doses of radiation.

Against this background, the [DSIWG](#) was established under the auspices of the [SCSC](#). The [DSIWG](#)’s aim is to develop clear, cross-sector guidance that reflects emerging best practice on how data (as opposed to software or hardware) should be managed in a safety-related context. For the most part, this guidance is based on well-established techniques, and it has been designed to be compatible with current safety standards and to integrate with existing safety management systems. What is new, however, is the explicit and relentless focus on data, making it a “first-class citizen” within system safety analyses. Because of this focus, this guidance should help organizations identify, analyse, evaluate and treat data-related risks, thus reducing the likelihood of data-related issues causing harm in the future.

Quick Start Guide

Data really powers everything that we do.

Jeff Weiner

This section provides a single-page introduction to [data safety guidance \(DSG\)](#). For first-time readers this should help place individual sections within an appropriate context. It should also help returning readers quickly navigate the document's contents.

- Systems are changing. The role of data is becoming more prominent. Hence, data needs to be considered as a “first-class citizen” in system safety analyses. This will help mitigate organizational and system-level risks associated with the use of data.
- A data safety management process has been developed. This is based on four phases:
 - establish context;
 - identify risks;
 - analyse risks; and
 - evaluate and [treat](#) risks.
- The underlying principles and an overview of the process are described in [chapter 2](#).
- Normative definitions and abbreviations are described in [chapter 3](#).
- The objectives associated with, and the outputs produced by, each phase are described in ??.
- The activities of each phase (and associated [tailoring information](#)) are described in ??.
- Additional guidance [information](#) for each phase is described in ??.
- A worked example is provided in ??.
- A collection of appendices provide more detail, including:
 - A discussion illustrating how the underlying principles link to the objectives (Appendix ??);
 - An [organizational data risk \(ODR\)](#) assessment questionnaire (Appendix ??);
 - A data safety culture questionnaire (Appendix ??);
 - A questionnaire to help assess the data maturity of a supplier (Appendix ??);
 - A list of data categories (Appendix ??);
 - A collection of [hazard and operability study \(HAZOP\)](#) guidewords (Appendix ??);
 - The suggested contents of a [data safety management plan \(DSMP\)](#) (Appendix ??);
 - A summary of accidents and incidents in which data was potentially a causal factor (Appendix ??);
 - A discussion of topics loosely related to system lifecycles (Appendix ??);
 - Considerations regarding [machine learning \(ML\)](#) (Appendix ??);
 - A discussion of the risks of AI and autonomy (Appendix ??);
 - An introduction to the concepts of both dark and dazzle data (Appendix ??);
 - The concepts of black swan, dragon king, perfect storm and Pudding Lane data (Appendix ??).
 - Considerations for the assurance and qualification of data-handling tools (Appendix ??).

- An introduction to the RADISH tool, that has been developed to assist in the application of the guidance within this document (Appendix ??).
- Issues that may arise when migrating, porting, importing or exporting data (Appendix ??).
- Some of the data issues that made management of the Covid-19 virus difficult (Appendix ??);
- Examples of ways that [data safety assurance levels \(DSALs\)](#) may be customised, with particular focus on likelihood (Appendix ??);
- Lists of acronyms, definitions and glossary entries ([Appendix 4](#)); and
- A collection of references ([Appendix 5](#)).

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1 Introduction

extract
principles

2 Principles (Normative)

Errors using inadequate data are much less than those using no data at all.
Charles Babbage

Hawkins *et al.* established some generic software safety assurance principles, which are commonly referred to as “4 + 1” [1]. Given the close links between software and data it is helpful to consider these principles from a data safety assurance perspective. The results are detailed below, with each principle being considered in turn.

2.1 Principle 1: data safety requirements shall be defined to address the data contribution to system hazards

Data pervades active system operation, as well as the system’s specification, realisation, [verification](#), [validation](#), certification, training, maintenance, and retirement. Moreover, data may be passed from one system to another, sometimes over a significant period of time. Data may be assimilated and converted from prior uses into new uses, or simply used as-is, by many systems. It is stored in media whose storage [integrity](#) decays. The system context for [data safety requirements](#) may be specific to a particular system’s or process’s use of the data, or it may be generalised to a class of related systems. Hence [data safety requirements](#) are needed for any safety-related system that interacts with data.

2.2 Principle 2: the intent of the data safety requirements shall be maintained throughout requirements decomposition

[Data safety requirements](#) establish the system’s safety properties for data, for the system’s use of data, for the management of data, and for the engineering lifecycle of both the system and its associated data. The system’s requirements hierarchy must preserve the intent of the [data safety requirements](#) (and hence the system’s safety-related [data properties](#)). Moreover, the applied engineering process for both the system’s realisation and subsequent lifecycle stages shall demonstrate that the data safety properties are preserved.

2.3 Principle 3: data safety requirements shall be satisfied

Evidence is required that the system satisfies all of the [data safety requirements](#) imposed on it for all anticipated operating conditions. Moreover, the [data safety requirements](#) that pertain to the data’s lifecycle outside of the system shall be evidentially demonstrated prior to the system acting on such data, or the system shall be able to adequately defend against unsatisfied [data safety requirements](#). In other words, either the data can be shown to demonstrate the required [data properties](#) prior to being used or the system can implement adequate defences and [mitigations](#) against data that does not conform to the required safety properties.

2.4 Principle 4: Hazardous system behaviour arising from the system's use of data shall be identified and mitigated

This is an intentionally broad statement because data is conceptual and not physical; it is the contextualised use of data that could result in a system hazard. Data safety assurance principle 1 deals with system-level hazards arising from data, whereas Data safety assurance principle 4 is concerned with hazards that arise from the way the system uses its data, that is, whether the system's design and implementation introduce further hazards. An example is a ship navigation system's display of hydrographic chart data, where a wide field display results in small features disappearing (due to image scale) when it is critical that situational awareness of such hazards is maintained. The associated objectives and outputs are shown in Part 2 Chapter 5.4.

2.5 Principle 5: The confidence established in addressing the data safety assurance principles shall be commensurate to the contribution of the data to system risk

(Called Principle 4+1 by Hawkins.) The confidence in the evidence that demonstrates establishment of the first four Data Safety Assurance Principles shall be proportionate to the contribution data (or a particular data artefact) makes to the system hazards.

expand
dark data,
make into
objective

Add
definitions
for
cygnology
types

Identify
process
for dark
data

3 Definitions (Normative)

I love data. I think it's very important to get it right, and I think it's good to question it.

Mary Meeker

artefact, data

An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

property, data

A characteristic that can be exhibited by a [data artefact](#).

safety assurance level, data

An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

safety requirement, data

A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

treatment

An action taken to reduce or control risk. This might be [mitigation](#) of the risk or elimination of the hazard.

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4 Acronyms, Definitions and Glossary (Discursive)

The plural of anecdote is not data.
Mark Berkoff

4.1 Acronyms

AI	artificial intelligence
DSAL	data safety assurance level
DSG	data safety guidance
DSIWG	Data Safety Initiative Working Group
DSMP	data safety management plan
HAZOP	hazard and operability study
LLM	large language model
ML	machine learning
ODR	organizational data risk
OED	Oxford English Dictionary
SCSC	Safety-Critical Systems Club
SSS	Safety-critical Systems Symposium

4.2 Definitions and Glossary

accuracy

- Closeness of agreement between a test result and the accepted reference value. Note that a test result can be observations or measurements. ISO 19113:2005 [8]
- A degree of conformance between the estimated or measured value and the true value. (EU) No 73/2010 [9]
- (Temporal) Correctness of the temporal references of an item (reporting of error in time measurement). Correctness of ordered events or sequences, if reported. [Validity](#) of data with respect to time. ISO 19138:2006 [10]

artefact, data

(Normative) An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

completeness

- Property of having every necessary part or element.
- Completeness of the data provided. RTCA/DO-200A

consistency, data

The property that the data adheres to a common world view (e.g., units).

correctness, data

self-[consistency](#), protection against alteration or corruption and [consistency](#) with the functional requirements of the [data-driven system](#). IEC 61508 Part 3 [3]

data

- A thing given or granted; something known or assumed as fact, and made the basis of reasoning or calculation; an assumption or premiss from which inferences are drawn. [Oxford English Dictionary \(OED\)](#)
- A reinterpretable representation of [information](#) in a formalized manner suitable for communication, interpretation or processing. ISO/IEC 2382 [14]

dataset

Identifiable collection of data. Note that a dataset may be a smaller grouping of data which, though limited by some constraint such as spatial extent or feature type, is located physically within a larger dataset. Theoretically, a dataset may be as small as a single feature or feature attribute contained within a larger dataset. A hardcopy map or chart may be considered a dataset. BS EN ISO 19131:2008 [15]

information

- Knowledge communicated concerning some particular fact, subject, or event; that of which one is apprised or told - intelligence, news - as contrasted with data. [OED](#)
- Knowledge that has a contextual meaning. ISO/IEC 2382 [14]

integrity, data

- The assurance that a data element retrieved from a storage system has not been corrupted or altered in any ways since the original data entry or latest authorised amendment. RTCA/DO-200A [17]
- The degree of assurance that a [data item](#) and its value have not been lost or altered since the data origination or authorised amendment. (EU) No 73/2010 [9]
- The degree of undetected (at system level) non-conformity of the input value of the [data item](#) with its output value. (EU) No 1207/2011 [18]
- The property of protecting the [accuracy](#) and [completeness](#) of assets, i.e., that which has value to the organization. ISO 27001:2013 [12]

item, data

Single attribute of a complete [dataset](#), which is allocated a value that defines its current status. (EU) No 73/2010 [9]

mitigation

Steps taken to control or prevent a hazard from causing harm and reduce risk to a tolerable or acceptable level. [19]

owner, data

(Normative) The individual or organization responsible for a particular [data artefact](#) or collection of [data artefacts](#).

property, data

(Normative) A characteristic that can be exhibited by a [data artefact](#).

response

(Normative) The way in which an identified risk is addressed; possible responses include avoid / eliminate, treat, or accept as sufficiently low.

safety assessment, data

(Normative) The process of explicitly considering data as part of a system safety assessment, via the means of [data artefacts](#), Data Properties and [DSALs](#).

safety assurance level, data

(Normative) An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

safety requirement, data

(Normative) A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

stakeholder

(Normative) An individual or organization that has some relationship to the system, possibly including a power of veto.

tailoring

Adaptation of processes etc. to be appropriate for a specific system and context

treat

To apply a [treatment](#)

treatment

(Normative) An action taken to reduce or control risk.

validation, data

- The activity whereby a data element is checked as having a value that is fully applicable to the identity given to the data element, or a set of data elements that is checked as being acceptable for their purpose. RTCA/DO-200A [17]
- Process of ensuring that data meets the requirements for the specified application or intended use. (EU) No 73/2010 [9]

validity, period of

Period between the date and time on which information becomes available and the date and time on which the [information](#) ceases to be effective. Based on (EU) No 73/2010 [9]

verification, data

Evaluation of the output of a process to ensure [correctness](#) and [consistency](#) with respect to the inputs and applicable data standards, rules and conventions used in that process. Based on (EU) No 73/2010 [9]

The normative list of definitions is at [chapter 3](#). Normative definitions have been repeated here for convenience.

5 References (Discursive)

Data opportunities multiply as the data is transformed.
Sun Tzu misquoted

Bibliography

- [1] R Hawkins, I Habli, and T Kelly. *The Principles of Software Safety Assurance*. International System Safety Society, Boston, Massachusetts, USA, 2013.
- [2] *SAE Aerospace Recommended Practice 4754A: Guidelines for development of civil aircraft and systems*. Society of Automotive Engineers, 2010.
- [3] Functional safety of electrical/electronic/ programmable electronic safety-related systems: Software requirements. Standard BS EN 61508-3:2010, International Electrotechnical Commission, June 2010.
- [4] Software Tool Qualification Considerations. Standard RTCA/DO-330, EUROCAE/ED-215, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, January 2012.
- [5] Die Lage der IT-Sicherheit in Deutschland 2014. URL <https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Publikationen/Lageberichte/Lagebericht2014.pdf>, December 2014. Accessed: 27 January 2021.
- [6] D LeBlanc and M Howard. *Writing secure code*. Microsoft Press, December 2002.
- [7] *Cyber Security and Safety Code of Practice*. IET, 2020. ISBN 978-1-83953-318-8.
- [8] Geographic Information. Quality Principles. Standard BS EN ISO 19113:2005, International Standards Organisation, 2005.
- [9] Commission Regulation (EU) No 73/2010 of 26 January 2010 laying down requirements on the quality of aeronautical data and aeronautical information for the single European sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:023:0006:0027:EN:PDF>, January 2010. Accessed: 27 January 2021.
- [10] Geographic Information. Data Quality Measures. Standard ISO/TS 19138:2006, International Standards Organisation, 2006.
- [11] Commission Regulation (EC) No 482/2008 of 30 May 2008 establishing a software safety assurance system, as amended. URL <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32008R0482&qid=1611781633022>, May 2008. Accessed: 27 January 2021, but no longer in force.
- [12] Information Technology — Security Techniques — Information Security Management Systems — Requirements. Standard BS ISO/IEC 27001:2013, International Standards Organisation, 2013.
- [13] Guidelines for ANS software safety assurance. Standard EUROCAE/ED-153, European Organisation for Civil Aviation Equipment, August 2009. Used for definitions only.
- [14] Information Technology. Vocabulary. Part 1: Fundamental terms. Standard ISO/IEC 2382-1:1993, International Standards Organisation, 1993.
- [15] Geographic Information. Data Product Specifications. Standard BS EN ISO 19131:2008, International Standards Organisation, 2008.

- [16] P Ensor. *Safety Analysis of Navigational Data*. University of York, September 2009.
- [17] Standards for processing aeronautical data. Standard RTCA/DO-200A, EUROCAE/ED-76, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, September 1998.
- [18] Commission Implementing Regulation (EU) No 1207/2011 of 22 November 2011 laying down requirements for the performance and the interoperability of surveillance for the single european sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2011:305:0035:0052:EN:PDF>, November 2011. Accessed: 27 January 2021.
- [19] Use of safety management systems by ATM service providers. Standard ESARR3, EUROCONTROL, 2011.

DATA IS HERE. DATA IS GROWING. DATA IS CAUSING HARM.

This book has been developed by the Safety-Critical Systems Club Data Safety Initiative Working Group (DSIWG) to provide guidance on how data, as distinct from hardware and software can be managed in a safety-related context.

"If you torture the data long enough, they confess – even to crimes that were never committed."

Nihat Bülent Gültekin

~

This is the seventh minor update since version 3.0. Paragraph numbering within the body of the document remains aligned with that major release. Thus users of any previous 3.x release of the guidance document will find migration to this edition takes little effort.

